

GroundedStyle: Selective Background Stylization

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Abstract

Neural Style Transfer (NST) is a popular image synthesis technique that combines the content of one image with the artistic style of another. However, conventional NST applies style uniformly across the entire image, which often alters the foreground subject and reduces visual clarity. In this project, we propose a segmentation-guided style transfer pipeline that applies artistic transformation only to the background while preserving the foreground object.

1. Introduction

Artistic image synthesis has gained significant attention in recent years due to its ability to combine the semantic structure of one image with the visual appearance of another. Among the most influential techniques in this domain is *Neural Style Transfer* (NST), which demonstrates how deep neural networks can be used to separate and recombine *content* and *style* representations of images. The classical NST formulation, as covered in the course notes of EE665 (Computer Vision and Deep Learning), serves as the starting point and inspiration for this project.

1.1. Motivation

Vanilla NST applies artistic style to the entire image uniformly. While this often produces visually pleasing outputs, it may also alter the foreground subject in undesirable ways, reducing clarity and disrupting the identity of the main object in the scene. In many practical image synthesis settings, it is preferable to stylize only the background while preserving the foreground structure and appearance. This motivates a more controllable style transfer pipeline that combines segmentation and generative modeling.

$$\mathcal{L}_{\text{NST}} = \alpha \mathcal{L}_{\text{content}} + \beta \mathcal{L}_{\text{style}} + \gamma \mathcal{L}_{\text{tv}}$$

where the generated image is optimized to preserve the content structure while matching the artistic style statistics.

1.2. Project Objective

The main objective of this project is to design a background-only style transfer framework in which the foreground remains largely unchanged while the background is transformed according to the desired artistic style. To achieve this, the project extends the idea of classical NST by introducing a segmentation-based masking stage followed by a learning-based stylization stage by training cycleGAN. The final goal is to obtain a system that can preserve subject identity while still generating visually rich and stylized outputs.

1.3. Segmentation Pipeline Design

To develop a robust segmentation-guided Neural Style Transfer system, multiple preprocessing pipelines were explored for reliable foreground-background separation. The effectiveness of the overall framework strongly depends on the quality of segmentation masks, as any inaccuracies directly affect the final stylized output.

We first experimented with pretrained *DeepLab* for segmentation. Although it provided reasonable results in some cases, the masks were not sufficiently reliable for general-purpose images, especially when the foreground object did not align well with the semantic categories learned by the model.

We then tried *YOLO*-based segmentation. However, its utility was limited by the fixed set of predefined classes, which made it less suitable for arbitrary images where the foreground object might not belong to one of the available categories. This restricted its applicability for generating reliable foreground masks across diverse images.

To address these limitations, the pipeline was refined by using *Grounding DINO* on the *COCO* dataset. In this setup, object names from COCO are used as prompts for segmentation, enabling more flexible object localization and mask generation.

1.4. Style Transfer Framework

For the style transfer component, we explored both a classical Neural Style Transfer (NST) approach and a generative modeling approach, with the latter forming the primary

contribution of this work.

An initial implementation was based on pretrained VGG19-based Neural Style Transfer. This method captures artistic style by matching feature correlations (Gram matrices) between a content image and a style image through iterative optimization. While effective in producing visually appealing results, NST applies stylization globally across the entire image. Even when combined with masking, the optimization process does not strictly constrain style application to the background, often leading to distortions in foreground regions.

To overcome these limitations, we adopt a GAN-based approach using a CycleGAN model trained for this task. Unlike NST, which relies on fixed feature representations, CycleGAN learns a mapping between two image domains. In our case, the model is trained to translate natural images into the artistic domain of Monet paintings using a dataset obtained from Kaggle.

Given an input image I , the generator produces a fully stylized output:

$$\hat{I} = G(I), \quad (1)$$

where $\hat{I}$ represents the Monet-style transformation.

To preserve the foreground, the final output is computed using mask-based blending:

$$I_{out} = M_{fg} \odot I + M_{bg} \odot \hat{I}, \quad (2)$$

where M_{fg} and M_{bg} denote the foreground and background masks, respectively.

This formulation ensures that stylization is restricted to the background region while maintaining the structural integrity of the foreground. Compared to VGG19-based NST, the GAN-based approach produces more consistent and realistic stylization by learning the underlying style distribution directly from data and enabling better spatial control during inference.

1.5. Contributions

The main contributions of this project are summarized as follows:

- A segmentation-guided style transfer framework that selectively applies artistic transformation to the background while preserving the foreground content.
- A systematic evaluation of multiple segmentation approaches, including DeepLab and YOLO, highlighting their limitations for general-purpose foreground extraction.
- An open-vocabulary, prompt-driven segmentation pipeline using Grounding DINO (with Grounded SAM refinement), where COCO object labels are used as textual prompts for flexible and robust mask generation.
- A background stylization approach that combines segmentation with a CycleGAN model trained on Monet

paintings (sourced from a Kaggle dataset), enabling realistic and data-driven artistic transformation.

- A comparative analysis between classical VGG19-based Neural Style Transfer and GAN-based stylization, demonstrating the advantages of learned style distributions and spatially controlled application.

2. Methodology

This section presents the proposed segmentation-guided stylization framework for selective background transformation. The primary objective is to preserve the foreground subject of an image while applying artistic stylization exclusively to the background. Unlike conventional Neural Style Transfer (NST), which applies style globally, our approach introduces spatial control through segmentation and combines it with both optimization-based and generative stylization techniques.

The overall framework consists of a baseline Neural Style Transfer (NST) pipeline for comparison and a proposed GAN-based approach using CycleGAN for learned stylization. Both pipelines share a common segmentation module for foreground-background separation.

2.1. Overview of the Proposed Framework

Given an input image I , the goal is to generate an output image I_{out} such that the foreground object remains unchanged while the background is stylized in the artistic domain of Monet paintings.

The pipeline consists of three main stages:

1. Foreground segmentation using a prompt-driven model.
2. Stylization of the image (either via NST or GAN).
3. Mask-based compositing to preserve the foreground.

A binary foreground mask M_{fg} is first obtained, and the background mask is defined as:

$$M_{bg} = 1 - M_{fg}. \quad (3)$$

These masks are used to control the spatial application of stylization.

2.2. Foreground-Background Segmentation

The success of the proposed method depends heavily on accurate segmentation. The objective is to isolate the primary subject of the image while ensuring that the remaining region can be safely stylized.

2.2.1. Initial Approaches

We first experimented with pretrained DeepLab models for semantic segmentation. Although DeepLab produces pixel-wise labels, its performance was inconsistent for general-purpose images. The generated masks were often coarse and failed to capture fine object boundaries, especially for objects outside common semantic categories.

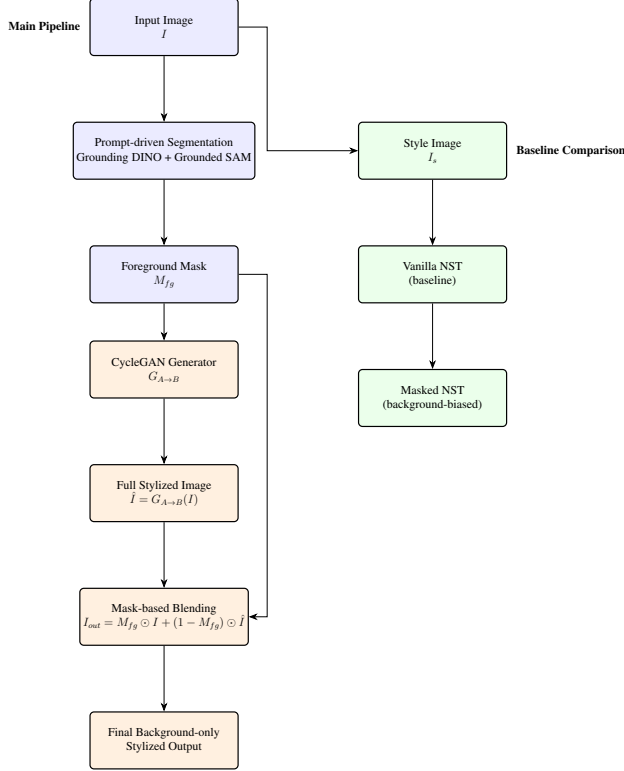


Figure 1. Pipeline of the proposed segmentation-guided background-only stylization framework.

We then explored YOLO-based segmentation. While YOLO offers efficient detection, it is constrained by a pre-defined set of object classes. This limitation made it unsuitable for arbitrary inputs where the foreground object may not belong to the available label space. Consequently, mask generation was unreliable in diverse scenarios.

2.2.2. Grounding DINO with COCO Prompts

To overcome the limitations of fixed-class segmentation models, we adopt Grounding DINO combined with segmentation refinement using Grounded SAM. This approach enables open-vocabulary object detection, allowing the model to localize objects based on natural language descriptions rather than predefined class labels.

In our implementation, object names derived from the COCO dataset are used as textual prompts to guide the segmentation process. This design allows the system to generalize across a wide range of images, as the foreground object can be specified dynamically using a descriptive label (e.g., “a person,” “a dog,” etc.).

Let p denote the textual prompt corresponding to the foreground object. The segmentation process can be formulated as:

$$M_{fg} = S(I, p), \quad (4)$$

where $S(\cdot)$ represents the prompt-driven segmentation

function.

Grounding DINO first identifies regions in the image that correspond to the given prompt, and Grounded SAM further refines these regions to produce accurate pixel-level masks. This combination results in high-quality segmentation that aligns well with object boundaries.

Compared to traditional approaches such as DeepLab and YOLO, this method provides significantly greater flexibility and robustness. Since the segmentation target is defined through prompts rather than fixed categories, the system can handle diverse and previously unseen objects. The resulting masks are therefore more precise and suitable for selective background stylization in the proposed pipeline.

2.3. Neural Style Transfer Baseline

As a baseline, we implemented a classical Neural Style Transfer pipeline using a pretrained VGG19 network. Given a content image I and a style image I_s , NST optimizes a generated image to minimize content and style losses:

$$\mathcal{L}_{total} = \alpha \mathcal{L}_{content} + \beta \mathcal{L}_{style}. \quad (5)$$

While this approach effectively captures artistic features through Gram matrix representations, it applies stylization uniformly across the entire image.

To introduce spatial control, we implemented a masked NST variant, where the stylization is biased toward the background using the segmentation mask. However, since NST relies on global feature statistics, it does not fully restrict style transfer to the background, often leading to partial distortion of foreground regions.

This limitation motivated the transition to a generative approach.

2.4. GAN-Based Stylization using CycleGAN

To achieve more controlled and realistic stylization, we adopt a CycleGAN-based framework. CycleGAN learns a mapping between two image domains without requiring paired data.

In this project, the two domains are:

- Domain A: natural images (input domain)
- Domain B: Monet-style paintings

The Monet dataset is obtained from a publicly available Kaggle dataset, which contains a collection of paintings in the style of Claude Monet. This dataset enables the model to learn the distribution of impressionist-style textures, colors, and brush strokes.

The CycleGAN consists of:

- Generators $G_{A \rightarrow B}$ and $G_{B \rightarrow A}$
- Discriminators D_A and D_B

The generator $G_{A \rightarrow B}$ learns to transform natural images into Monet-style images:

$$\hat{I} = G_{A \rightarrow B}(I). \quad (6)$$

2.5. Training Objective

CycleGAN is trained using a combination of adversarial and cycle-consistency losses.

The adversarial loss for domain B is:

$$\mathcal{L}_{adv}^B = \mathbb{E}_{I_B}[\log D_B(I_B)] + \mathbb{E}_{I_A}[\log(1 - D_B(G_{A \rightarrow B}(I_A)))]. \quad (7)$$

To enforce consistency between domains, the cycle-consistency loss is used:

$$\mathcal{L}_{cycle} = \|G_{B \rightarrow A}(G_{A \rightarrow B}(I_A)) - I_A\|_1. \quad (8)$$

The total loss is a weighted combination:

$$\mathcal{L}_{total} = \mathcal{L}_{adv} + \lambda_{cycle} \mathcal{L}_{cycle}. \quad (9)$$

This training enables the model to learn a realistic Monet-style transformation.

2.6. Inference and Background-Only Stylization

During inference, the trained CycleGAN model is applied to the entire input image to produce a fully stylized version:

$$\hat{I} = G_{A \rightarrow B}(I). \quad (10)$$

However, to preserve the foreground, the final output is constructed using mask-based blending:

$$I_{out} = M_{fg} \odot I + M_{bg} \odot \hat{I}. \quad (11)$$

This ensures that:

- The foreground region remains identical to the original image.
- The background region is stylized using the learned Monet representation.

2.7. Implementation Details

The overall framework consists of a baseline Neural Style Transfer (NST) pipeline and a proposed segmentation-guided GAN-based stylization pipeline. The NST-based approach is used for comparison, while the GAN-based pipeline represents the primary contribution of this work.

Baseline NST-based pipeline:

- Input image and style image are loaded.
- Foreground mask is generated using Grounded SAM.
- Vanilla NST is applied to the full image.
- Masked NST is applied to emphasize background stylization.

Proposed CycleGAN-based pipeline:

- Foreground mask is generated using prompt-based segmentation.
- The CycleGAN model (trained on Monet dataset) is applied to the image.
- The stylized output is blended with the original image using the segmentation mask.

3. Results

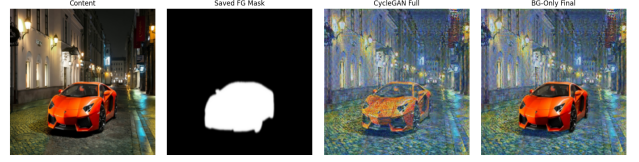


Figure 2. Original Image



Figure 3. Foreground Mask



Figure 4. CycleGAN applied to the entire image



Figure 5. Final fully processed image

The above figures present a qualitative comparison of the proposed background-only stylization approach. The first image shows the original content image, followed by the generated foreground mask obtained using the segmentation module. The mask accurately captures the primary object (car), effectively separating it from the background. The third image shows the output of the CycleGAN model applied to the entire image. As expected, the stylization is applied globally, affecting both the foreground and background. While the artistic transformation is visually appealing, the foreground object is distorted and loses its original clarity. The final image demonstrates the result of the proposed method. Using mask-based compositing, the stylized background is combined with the original foreground. This preserves the structural integrity, color, and details of the foreground object while applying Monet-style transformation exclusively to the background. The comparison highlights the effectiveness of the proposed approach in achieving spatially controlled stylization. Unlike conventional methods, the foreground remains visually consistent and the background exhibits strong artistic characteristics, resulting in a more coherent and aesthetically pleasing output.

4. Novelty of the Research

The proposed work introduces a novel approach to Neural Style Transfer by incorporating segmentation-driven spatial control into the stylization process. While traditional style transfer methods apply artistic transformations uniformly across the entire image, this work focuses on selective stylization, specifically targeting background regions while preserving foreground content.

A key novelty of this research lies in the integration of open-vocabulary segmentation with generative style transfer. By leveraging Grounding DINO combined with Grounded SAM, the framework enables prompt-driven foreground extraction using natural language descriptions. This overcomes the limitations of conventional segmentation methods that rely on fixed class labels and improves adaptability to diverse, real-world images.

Another major contribution is the use of a CycleGAN model trained specifically for this task using a Monet-style dataset obtained from Kaggle. Unlike classical VGG19-based Neural Style Transfer, which relies on feature-space optimization, the trained CycleGAN learns the underlying distribution of artistic styles directly from data. This enables the generation of more consistent, realistic, and high-quality stylized outputs.

A distinctive aspect of the proposed framework is the decoupling of stylization learning and spatial control. The CycleGAN is trained on full-image domains to learn style translation, while segmentation-based masking is applied during inference to restrict stylization to the background. This design eliminates the need for training on masked datasets while still achieving precise region-specific control.

Furthermore, the work includes a comparative study between classical NST and GAN-based approaches, highlighting the limitations of optimization-based methods and demonstrating the effectiveness of generative models for controlled stylization.

Overall, the novelty of this research lies in the unified integration of:

- Prompt-driven, open-vocabulary segmentation,
- Custom-trained CycleGAN for Monet-style transformation,
- Mask-based compositing for spatially controlled stylization.

5. Conclusion

In this project, we addressed a key limitation of conventional Neural Style Transfer (NST), namely its inability to control where stylization is applied within an image. Standard NST methods apply artistic transformations uniformly, often distorting important foreground subjects. To overcome this, we proposed a segmentation-guided stylization framework that selectively applies style to the background while preserving the foreground.

We explored multiple segmentation approaches, including DeepLab and YOLO, and identified their limitations in handling diverse, real-world images. To achieve more robust and flexible foreground extraction, we adopted a prompt-driven approach using Grounding DINO combined with Grounded SAM, enabling open-vocabulary segmentation through natural language prompts.

For the stylization component, we compared classical VGG19-based NST with a generative approach using CycleGAN. While NST provided a useful baseline, it lacked precise spatial control and often affected the foreground despite masking. In contrast, the CycleGAN-based approach, trained on Monet-style paintings, was able to learn the underlying artistic distribution and produce more consistent and realistic stylizations.

By combining segmentation with GAN-based stylization and mask-based compositing, the proposed framework achieves effective background-only stylization while preserving foreground integrity. The results demonstrate that incorporating spatial awareness into style transfer significantly improves visual quality and usability.

Future work may explore improving segmentation accuracy further, integrating more advanced generative models, and extending the framework to support multiple objects or region-specific stylization.

Appendix

This section contains the project repository and dataset links used in this work.

Project Repository

- [Link to Zip](#)
- [GitHub Repository](#)

Dataset Links

- **Dataset 1:** [COCO 2017 Dataset](#)
- **Dataset 2:** [Monet2Photo Dataset](#)

6. References

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